

Post-WKB Mathematical Derivations for the SU(2) Yang–Mills Kelvin–Helmholtz Program

A complete, step-by-step record of every derivation developed after `khaxn.pdf` and `wkb.pdf`

YM–KH simulation program

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Abstract

This document collects, in full step-by-step detail, all mathematical derivations produced by the SU(2) Yang–Mills Kelvin–Helmholtz (KH) program *after* the two foundational notes: `khaxn.pdf` (the linearized Yang–Mills–Wong system in the $a_x = 0$ gauge and the decoupled a_z wave equation) and `wkb.pdf` (the $\mathcal{O}(v^2)$ reduction to a Schrödinger form and the harmonic-core quantization, its eq. (33)). Results of those two notes are used *as given*; everything else is derived here from first principles. The main chain of results is: the quartic dispersion polynomial and its $k_z = 0$ chromo-Weibel limit (§2); the time-domain model as simulated, its Wong precession source and its exact energy budget including the frozen-background “battery” term (§3); the linearized six-field circular system (§4); the local dispersion relation, the tachyonic (Nielsen–Olesen–class) outer branch, and the Airy wall estimate (§5); the exact reduction of the six-field system to a single second-order ODE (§6); the precession-resonance drive, the hard growth ceiling $\gamma^3 \leq \alpha V_0^2$ and the natural units of the shear branch (§7); the quantization theory (models A and B) (§8); the confinement-set peak formula $k_z^{\text{peak}} \approx \alpha V_0 (\xi_w - \ln 2) + c (\alpha/V_0)^{1/3}$ and the windowed roll-off (§9); the shear-width (quantization budget) scaling (§10); the diagnosis of the WKB quartic’s domain of validity (§11); the cold two-stream rates in code units (§12); the nonlinear cavitation threshold and the quadratic back-reaction law (§13); the warm isothermal closure (§14); and the eigenmode-seeding theory (§15). Appendices give the helicity-projection identities, the filter/sponge operators, and the numerical validation ledger.

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1 Introduction, given results, and conventions

1.1 What is taken as given

Two documents predate everything here and their results are used without re-derivation.

Given 1: the linearized Yang–Mills–Wong system (khaxn.pdf). Working in the $a_x^a = 0$ gauge with a background flow $\vec{V}_0 = V_{0z}(x)\hat{z}$, background color charge $Q_0^a = \delta^{a1}$, and color-1 background potentials $A_{0z}^1(x)$, $\phi_0^1(x)$, that note derives (its eqs. (1)–(21)) the perturbed magnetic-field components, the perturbed color charges $q^{2,3}$, and the linearized Ampère equations, and closes them (its eqs. (22)–(23)) into decoupled wave equations for the circular combinations $a_y^\pm$, $a_z^\pm$. In particular its eq. (23),

$$(-\partial_x^2 - \omega^2 + k_z^2 + \alpha^2(\phi_0^1)^2 - \alpha^2(A_{0z}^1)^2) a_z^+ = \frac{\alpha^2 V_{0z}}{\Omega + \alpha \Phi_0^1} \left(V_{0z} - \frac{k_z + \alpha A_{0z}^1}{\omega + \alpha \phi_0^1} \right) a_z^+, \quad (1.1)$$

with $\Omega = \omega - k_z V_{0z}$ and $\Phi_0^1 = V_{0z} A_{0z}^1$, is the frequency-domain ancestor of everything in §6 below; we will meet the combinations $k_z \pm \alpha A_{0z}^1$ and the beam resonance denominator $\Omega + \alpha \Phi_0^1$ again, derived independently from the time-domain system.

Given 2: the harmonic-core WKB quantization (wkb.pdf). Expanding the step-potential form of (1.1) to $\mathcal{O}(v^2)$ produces a Schrödinger form $\partial_x^2 a_z + Q_\pm(x) a_z = 0$ with

$$Q_\pm(x) = Q_0^\pm - |C_1|x^2 + C_2x^4, \quad |C_1| = \frac{\alpha^3 v^2}{2\omega^2}, \quad C_2 = \frac{\alpha^2 v^2}{4}, \quad (1.2)$$

$$Q_0^\pm = Q_{0,\text{mean}} \pm \delta, \quad Q_{0,\text{mean}} = \omega^2 - k_z^2 - \frac{\alpha^2 v k_z}{\omega^2}, \quad \delta = \frac{\alpha^2 v^2}{\omega} \left(1 - \frac{k_z^2}{\omega^2} \right). \quad (1.3)$$

The Bohr–Sommerfeld condition applied to the harmonic core, with the $\pm\delta$ jump cancelling between the two half-integrals, gives the exact quantization of the harmonic core (wkb.pdf eq. (33)):

$$\boxed{Q_{0,\text{mean}} = (2n + 1)\sqrt{|C_1|}} \quad n = 0, 1, 2, \dots \quad (1.4)$$

Everything downstream of (1.4) — the quartic polynomial, its $k_z = 0$ limit, its regime of validity, and its eventual replacement by the exact-action theory — is derived in this document.

1.2 Units, geometry, and notation

Code units are used throughout: $c = \varepsilon_0 = 1$; each beam has unperturbed density $n_0 = 1$, particle mass $m = 1$ and color-charge magnitude $|Q^1| = 1$, so that *each beam's* plasma frequency is

$$\omega_p^2 = \frac{n_0 Q^2}{m \varepsilon_0} = 1. \quad (1.5)$$

Time is measured in these plasma units (“TU”). The simulation domain is the periodic box $L_x = 6\pi$, $L_z = 2\pi$, so an integer mode index k in z has $k_z = k$ exactly. The shear layer sits at $x = L_x/2$ and all x -profiles are written in the stretched coordinate

$$\xi = \frac{x - L_x/2}{\varepsilon_s}, \quad (1.6)$$

where ε_s (called EPS in the code, typically 0.15) is the shear half-width. The two cold beams A and B counter-stream in z and carry opposite color-1 charges:

$$v_{zA} = +V_0 \tanh \xi, \quad Q_A^1 = +1; \quad v_{zB} = -V_0 \tanh \xi, \quad Q_B^1 = -1. \quad (1.7)$$

The frozen color-1 background potential of the production (mode 1/6) geometry is

$$A_{z1}(\xi) = -V_0 \ln \cosh \xi, \quad \text{and we define} \quad u(\xi) \equiv -\alpha A_{z1}(\xi) = \alpha V_0 \ln \cosh \xi \geq 0. \quad (1.8)$$

For $|\xi| \gtrsim 2$, $\ln \cosh \xi = |\xi| - \ln 2 + \mathcal{O}(e^{-2|\xi|})$, so

$$u(\xi) \simeq \alpha V_0 (|\xi| - \ln 2) \quad (|\xi| \gtrsim 2). \quad (1.9)$$

$\alpha \equiv \alpha_{\text{YM}}$ is the coupling. Perturbations grow as $e^{\gamma t}$; the frequency-domain results of the two given notes map onto this via

$$\omega = i\gamma, \quad \omega^2 = -\gamma^2. \quad (1.10)$$

Two derived scales recur throughout (§7):

$$\Gamma \equiv (\alpha V_0^2)^{1/3} \text{ (rate scale / ceiling),} \quad K \equiv \left(\frac{\alpha}{V_0}\right)^{1/3} \text{ (wavenumber scale),} \quad \Gamma = V_0 K. \quad (1.11)$$

Hats denote quantities measured in these scales: $\gamma = \Gamma \hat{\gamma}$, $k_z = K \hat{k}$, etc. A window (sponge or hard cut) of radius ξ_w (in ξ -units) limits the mode region in production runs.

1.3 Reading map

Sections 2–15 are ordered logically, not chronologically: model $\rightarrow$ linearization $\rightarrow$ local theory $\rightarrow$ exact reduction $\rightarrow$ drive/ceiling $\rightarrow$ quantization $\rightarrow$ dispersion-curve consequences $\rightarrow$ scalings $\rightarrow$ nonlinear/auxiliary theory. Each section states which numerical validation anchors it; the ledger is collected in Appendix C.

2 From the harmonic-core quantization to the quartic, and the $k_z = 0$ chromo-Weibel rate

2.1 Clearing eq. (33) to a quartic polynomial

Step 2.1.1. Substitute the definitions (1.2)–(1.3) into (1.4), with the background velocity scale written $v = V_0$:

$$\omega^2 - k_z^2 - \frac{\alpha^2 V_0 k_z}{\omega^2} = (2n+1) \sqrt{\frac{\alpha^3 V_0^2}{2\omega^2}}. \quad (2.1)$$

Step 2.1.2. The square root on the right is resolved as $\sqrt{\alpha^3/2} V_0/\omega$. For real positive ω this is elementary; for the complex (growing) roots we are about to find, this choice of branch defines the analytic continuation used everywhere in the program (the continuation is fixed by requiring the $k_z = 0$ limit below to reproduce the direct cubic factorization, which it does). Define the constant

$$C \equiv (2n+1) \sqrt{\frac{\alpha^3}{2}} V_0. \quad (2.2)$$

Then (2.1) reads

$$\omega^2 - k_z^2 - \frac{\alpha^2 V_0 k_z}{\omega^2} = \frac{C}{\omega}. \quad (2.3)$$

Step 2.1.3. Multiply through by ω^2 (legitimate: $\omega = 0$ is not a root of (2.1)) and collect all terms on one side:

$$\boxed{\omega^4 - k_z^2 \omega^2 - C \omega - \alpha^2 V_0 k_z = 0} \quad (2.4)$$

This is the *WKB quartic* (also `wkbfull.pdf` eq. (4)); the fastest-growing root $\gamma = \max \text{Im } \omega$ over its four roots is the “ γ_{WKB} ” quoted in all program tables.

Remark 2.1 (The growing root is not purely imaginary away from $k_z = 0$). An early implementation assumed $\text{Re } \omega = 0$ and bisected on real γ . Substituting $\omega = i\gamma$ into (2.4) gives $\gamma^4 + k_z^2 \gamma^2 - \alpha^2 V_0 k_z = iC\gamma$, whose left side is real and right side imaginary — so a purely imaginary root exists only where $C\gamma$ can vanish, i.e. at $k_z = 0$ (where the real part balances differently, see below) or in the trivial limit $C \rightarrow 0$. Numerically the purely-imaginary assumption erred by up to $\sim 50\%$ at $k_z = 0.25$ and $\sim 10\%$ at $k_z = 1$, shrinking with k_z ; the full complex quartic is solved everywhere in current tooling.

2.2 The $k_z = 0$ limit: chromo-Weibel growth rate

Step 2.2.1. Set $k_z = 0$ in (2.4). Both the $k_z^2 \omega^2$ term and the constant term $\alpha^2 V_0 k_z$ vanish:

$$\omega^4 - C\omega = 0 \quad \Longleftrightarrow \quad \omega(\omega^3 - C) = 0. \quad (2.5)$$

Step 2.2.2. Discard the root $\omega = 0$ (no dynamics). The cubic factor gives the three cube roots of the positive real number C :

$$\omega \in \left\{ C^{1/3}, C^{1/3} e^{2\pi i/3}, C^{1/3} e^{-2\pi i/3} \right\}. \quad (2.6)$$

Step 2.2.3. The growing root is the one with positive imaginary part, $\omega = C^{1/3} e^{2\pi i/3}$, with

$$\gamma = \text{Im } \omega = C^{1/3} \sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2} C^{1/3}. \quad (2.7)$$

Step 2.2.4. Insert C from (2.2) for the fundamental $n = 0$:

$$\boxed{\gamma_{\text{Weibel}}(k_z=0) = \left(\sqrt{\frac{\alpha^3}{2}} V_0 \right)^{1/3} \frac{\sqrt{3}}{2} = \left(\sqrt{\frac{\alpha^3}{2}} V_0 \right)^{1/3} \sin 60^\circ} \quad (2.8)$$

Remark 2.2 (Physical content and validation). At $k_z = 0$ the unstable channel is a color-mixing filamentation (“chromo-Weibel”) mode: the background A_{z1} mixes the color-2 and color-3 electromagnetic fields through the non-Abelian Ampère term $-\alpha(A_z \times B_y)^a$ and Faraday term $+\alpha(A_z \times E_x)^a$ (§3), producing z -uniform exponentially growing color-2/3 fields. Direct simulation reproduced (2.8) to 0.5%, the strongest single anchor of the quartic. This mode is deliberately suppressed in production KH campaigns (`suppress_kz0=1` plus hyperdiffusion, Appendix B); at the reference point of the self-consistency study (§13.2), $\alpha = 1$, $V_0 = 0.05$, formula (2.8) gives $\gamma = 0.284$, exactly the measured irreducible $k_z = 0$ growth there.

3 The time-domain model as posed

The nonlinear system integrated by the CUDA code is the time-domain counterpart of the linearized system of `khaxn.pdf`. It is stated here as the model definition; the derivations in this section are (i) the Wong-equation origin of the precession source, (ii) the invariant $B_y^a = -\partial_x A_z^a$ of the potential/Faraday pair, and (iii) the exact energy budget, which identifies the frozen background as an external battery.

3.1 Model equations

Fields: for each color $a \in \{1, 2, 3\}$, the 2D (x, z) fields $E_x^a, E_z^a, B_y^a, A_z^a$; for each beam $s \in \{A, B\}$, the fluid fields n_s, p_{xs}, p_{zs} (momentum densities; $v_{is} = p_{is}/n_s$) and the color-charge densities Q_s^a . The gauge is $A_x^a = A_y^a = 0$ with only A_z^a retained ($\partial_y \equiv 0$, no y -components of E or of the fluid velocity are excited by the dynamics below).

Fluid sector (each beam; FCT advection):

$$\partial_t n + \nabla \cdot (n \vec{v}) = 0, \quad (3.1)$$

$$\partial_t p_x + \nabla \cdot (\vec{v} p_x) = F_x = -n \sum_a Q^a (E_x^a + v_z B_y^a), \quad (3.2)$$

$$\partial_t p_z + \nabla \cdot (\vec{v} p_z) = F_z = -n \sum_a Q^a (E_z^a - v_x B_y^a), \quad (3.3)$$

$$\partial_t Q^a + \nabla \cdot (\vec{v} Q^a) = \alpha v_z (\vec{Q} \times \vec{A}_z)^a, \quad (3.4)$$

where $(\vec{Q} \times \vec{A}_z)^a = \epsilon^{abc} Q^b A_z^c$ is the SU(2) adjoint cross product. (In (3.4) the Q^a are advected as densities; combined with (3.1) this is equivalent to the advective form $dQ^a/dt = \alpha v_z (\vec{Q} \times \vec{A}_z)^a$ along fluid trajectories, which is what Wong's equation supplies, §3.2.)

Field sector (leapfrog on a Yee stagger):

$$\partial_t E_x^a = -\partial_z B_y^a - \alpha (\vec{A}_z \times \vec{B}_y)^a - J_x^a, \quad (3.5)$$

$$\partial_t E_z^a = +\partial_x B_y^a - J_z^a, \quad (3.6)$$

$$\partial_t B_y^a = \partial_x E_z^a - \partial_z E_x^a + \alpha (\vec{A}_z \times \vec{E}_x)^a, \quad (3.7)$$

$$\partial_t A_z^a = -E_z^a \quad (A_z^1 \text{ held fixed when } \texttt{freeze_az1}=1), \quad (3.8)$$

$$J_i^a = -n_A Q_A^a v_{iA} - n_B Q_B^a v_{iB}, \quad i \in \{x, z\}. \quad (3.9)$$

Equation (3.6) carries no non-Abelian term: the candidate commutator $\alpha \epsilon^{abc} A_z^b F^{zz,c}$ vanishes identically because only the z -component of the potential is kept. The sign conventions (charge enters the current and the force with an overall minus) are the model's own; their only invariant consequences are the products current $\times$ field and force $\times$ velocity, and those are fixed by the energy budget below.

3.2 Derivation of the precession source from Wong's equations

Step 3.2.1. Wong's equation for the color charge of a classical particle moving on a worldline $x^\mu(\tau)$ in a background potential A_μ^b is

$$\frac{dQ^a}{d\tau} + g \epsilon^{abc} \frac{dx^\mu}{d\tau} A_\mu^b Q^c = 0. \quad (3.10)$$

Step 3.2.2. Non-relativistic limit ($\tau \rightarrow t$, $dx^\mu/d\tau \rightarrow (1, \vec{v})$), temporal gauge $A_0^b = 0$, and only $A_z^b \neq 0$:

$$\frac{dQ^a}{dt} = -g \epsilon^{abc} v_z A_z^b Q^c. \quad (3.11)$$

Step 3.2.3. Rename $g \rightarrow \alpha$ and swap the summed indices $b \leftrightarrow c$ using antisymmetry, $\epsilon^{abc} A^b Q^c = -\epsilon^{abc} Q^b A^c$:

$$\frac{dQ^a}{dt} = +\alpha v_z \epsilon^{abc} Q^b A_z^c = \alpha v_z (\vec{Q} \times \vec{A}_z)^a, \quad (3.12)$$

which is exactly the source in (3.4). Because $|\vec{Q}|$ is conserved by a cross-product rotation, $Q_{A,B}^1 = \pm 1$ remains a unit adjoint vector until advection mixes fluid elements.

3.3 The invariant $B_y^a = -\partial_x A_z^a$ of the potential pair

From (3.8), $\partial_t(-\partial_x A_z^a) = \partial_x E_z^a$. Comparing with (3.7), the difference field $W^a \equiv B_y^a + \partial_x A_z^a$ obeys

$$\partial_t W^a = -\partial_z E_x^a + \alpha (\vec{A}_z \times \vec{E}_x)^a. \quad (3.13)$$

Thus $B_y = -\partial_x A_z$ (the magnetostatic relation, matching `khaxn.pdf` eqs. (4)–(6), $b_y^a = -\partial_x a_z^a$) is preserved exactly for z -uniform, color-aligned configurations (e.g. the color-1 background itself, for which the right side of (3.13) vanishes), but *not* for the color-2/3 wave sector, where B_y carries the additional dynamics injected through E_x . This is why the code treats B_y^a as an independent evolved field, and why a current-consistent B_{y1} initialization (`init_by1_eq`) is needed when the color-1 sector is live (§13.2).

3.4 Exact energy budget: conservative core plus the frozen-background battery

The monitored energy is

$$E(t) = \int dx dz \left[\frac{1}{2} \sum_a ((E_x^a)^2 + (E_z^a)^2 + (B_y^a)^2) + \sum_{s=A,B} \frac{p_{xs}^2 + p_{zs}^2}{2n_s} \right]. \quad (3.14)$$

We differentiate term by term using (3.1)–(3.9), at the continuum level and over the periodic box (so all pure divergences integrate to zero).

Step 3.4.1 (Transport terms). The Maxwell curl terms combine into divergences:

$$E_x^a(-\partial_z B_y^a) + B_y^a(-\partial_z E_x^a) = -\partial_z(E_x^a B_y^a), \quad E_z^a(\partial_x B_y^a) + B_y^a(\partial_x E_z^a) = \partial_x(E_z^a B_y^a), \quad (3.15)$$

i.e. a Poynting flux; both integrate to zero. Likewise, for smooth flow the FCT advection terms transport $\frac{1}{2}p^2/n$ conservatively: using (3.1)–(3.3) without sources, $\partial_t(p_i^2/2n) + \nabla \cdot (\vec{v} p_i^2/2n) = 0$.

Step 3.4.2 (Current work cancels against the Lorentz work). The field sector loses $\sum_a (J_x^a E_x^a + J_z^a E_z^a) = \vec{J}^a \cdot \vec{E}^a$; with (3.9) this equals $-\sum_s n_s \sum_a Q_s^a \vec{v}_s \cdot \vec{E}^a$. The fluid sector gains $\sum_s \vec{v}_s \cdot \vec{F}_s$ with $\vec{F}_s$ from (3.2)–(3.3); the magnetic parts cancel within the dot product,

$$v_x(v_z B_y^a) + v_z(-v_x B_y^a) = 0, \quad (3.16)$$

leaving $\sum_s \vec{v}_s \cdot \vec{F}_s = -\sum_s n_s \sum_a Q_s^a \vec{v}_s \cdot \vec{E}^a$. The two contributions are equal and opposite: energy flows between fields and fluid with no loss. (This is the invariant content of the model’s charge-sign convention: the same minus sign appears in (3.9) and in (3.2)–(3.3), so all work terms pair correctly.)

Step 3.4.3 (Non-Abelian cross terms: the battery). The remaining contributions are the two commutator terms,

$$\left. \frac{dE}{dt} \right|_{\text{NA}} = \int \left[-\alpha \sum_a E_x^a (\vec{A}_z \times \vec{B}_y)^a + \alpha \sum_a B_y^a (\vec{A}_z \times \vec{E}_x)^a \right]. \quad (3.17)$$

Write both as scalar triple products in color space, $[\vec{X}, \vec{Y}, \vec{Z}] \equiv \epsilon^{abc} X^a Y^b Z^c$. The triple product is odd under exchanging its first and third arguments, so $[\vec{B}_y, \vec{A}_z, \vec{E}_x] = -[\vec{E}_x, \vec{A}_z, \vec{B}_y]$ and

$$\left. \frac{dE}{dt} \right|_{\text{NA}} = -2\alpha \int [\vec{E}_x, \vec{A}_z, \vec{B}_y] = -2\alpha \int \vec{A}_z \cdot (\vec{B}_y \times \vec{E}_x). \quad (3.18)$$

This does *not* vanish in general: the model’s quadratic energy (3.14) is not the full Yang–Mills energy (which hides the interaction energy inside the field-strength commutators), and (3.18) is precisely the exchange with the potential sector.

Step 3.4.4 (Evaluation on the frozen background). With $\vec{A}_z = (A_{z1}, 0, 0)$ plus small wave components, the leading term of (3.18) is

$$\left. \frac{dE}{dt} \right|_{\text{NA}} = -2\alpha \int A_{z1} (B_y^2 E_x^3 - B_y^3 E_x^2) = -2\alpha \int A_{z1} \text{Im}(b^* e_x), \quad (3.19)$$

where $b = B_y^2 + iB_y^3$ and $e_x = E_x^2 + iE_x^3$ are the circular pairs of §4. For any mode obeying the linearized Ampère- x relation $\gamma e_x = -i\Omega_A b$ (eq. (4.5) below), $\text{Im}(b^* e_x) = -\Omega_A |b|^2 / \gamma$, so

$$\left. \frac{dE}{dt} \right|_{\text{NA}} = \frac{2\alpha}{\gamma} \int A_{z1} \Omega_A |b|^2. \quad (3.20)$$

In the tachyonic outer region (§5) $\Omega_A = k_z - u < 0$ while $A_{z1} < 0$: the integrand is positive — the frozen background *feeds* the wave, and with `freeze_az1=1` the battery is bottomless. In the shear-branch well $\Omega_A > 0$ and the same term is an energy drain; the shear branch instead grows at the expense of the beams’ kinetic energy through the (internally cancelling) current-work channel. Equation (3.20) is the energetic statement of the freeze artifact discussed in §5.2 and measured in §13.2.

Remark 3.1 (Covariant Gauss constraint). The monitored (not enforced) constraint is the covariant Gauss law $\partial_x E_x^a + \partial_z E_z^a + \alpha \epsilon^{abc} A_z^b E_z^c = \rho^a$ with $\rho^a = -\sum_s n_s Q_s^a$, checked numerically by `gauss_check.py`; the leapfrog scheme preserves it to discretization accuracy in the same way the standard Yee scheme preserves the Abelian constraint.

4 Linearization: the six-field circular system

This section derives, in full detail, the linear eigenvalue system implemented by the 1D solver `ym_eigenmode.py` and reduced by all later theory. It was validated against the 2D simulation at the 1–4% level across the production grid.

4.1 Background and perturbation ansatz

Background (mode 1/6 geometry): beams (1.7), frozen potential (1.8), all other fields zero. Write $v(\xi) \equiv V_0 \tanh \xi$ so $v_{zA} = v$, $v_{zB} = -v$. Perturbations of the color-2/3 sector are combined into complex “circular” pairs

$$b = B_y^2 + iB_y^3, \quad e_x = E_x^2 + iE_x^3, \quad e_z = E_z^2 + iE_z^3, \quad a = A_z^2 + iA_z^3, \quad q_{A,B} = Q_{A,B}^2 + iQ_{A,B}^3, \quad (4.1)$$

each with the ansatz $f(x, z, t) = \hat{f}(x) e^{ik_z z + \gamma t}$, i.e.

$$\partial_t \rightarrow \gamma, \quad \partial_z \rightarrow ik_z. \quad (4.2)$$

Why circular pairs work: every non-Abelian term in (3.5)–(3.8) and (3.4) is a cross product with the color-1 background, and for any color vector $\vec{X} = (X^1, X^2, X^3)$ the 2/3-components of $\vec{A}_z \times \vec{X}$ with $\vec{A}_z = (A_{z1}, 0, 0)$ background are

$$(\vec{A}_z \times \vec{X})^2 = -A_{z1} X^3, \quad (\vec{A}_z \times \vec{X})^3 = +A_{z1} X^2 \implies (\vec{A}_z \times \vec{X})^2 + i(\vec{A}_z \times \vec{X})^3 = i A_{z1} X^c, \quad (4.3)$$

where $X^c = X^2 + iX^3$. (Contributions with the perturbed a against background color-2/3 fields vanish because the latter are zero; the color-1 wave fields B_y^1, E_x^1 couple only at second order or through backgrounds excluded from this model — see the scope remark in §4.4.) A background cross product therefore acts on a circular pair as multiplication by iA_{z1} : color rotation becomes a $U(1)$ phase, and the 12 real color-2/3 fields close into 6 complex ones.

Define the two shifted wavenumbers that this produces,

$$\boxed{\Omega_A(\xi) = k_z + \alpha A_{z1} = k_z - u(\xi), \quad \Omega_F(\xi) = k_z - \alpha A_{z1} = k_z + u(\xi).} \quad (4.4)$$

4.2 Derivation of the six equations

Step 4.2.1 (Ampère- x). Linearize (3.5) for colors 2,3 and combine as (2) + i(3). The curl term gives $-ik_z b$. The commutator, by (4.3) with $\vec{X} = \vec{B}_y$, gives $-\alpha i A_{z1} b$. The current perturbation vanishes: by (3.9), $\delta J_x^{2,3}$ needs either a background v_x (zero) or a product of two perturbations ($Q_s^{2,3}$ and δv_{xs}). Hence

$$\gamma e_x = -ik_z b - i\alpha A_{z1} b = -i\Omega_A b. \quad (4.5)$$

Step 4.2.2 (Faraday). Linearize (3.7): $\partial_x E_z \rightarrow e'_z$, $-\partial_z E_x \rightarrow -ik_z e_x$, and the commutator $+\alpha(\vec{A}_z \times \vec{E}_x) \rightarrow +i\alpha A_{z1} e_x$:

$$\gamma b = -ik_z e_x + i\alpha A_{z1} e_x + e'_z = -i\Omega_F e_x + e'_z. \quad (4.6)$$

Note the opposite relative sign of the two αA_{z1} shifts in (4.5) vs (4.6): Ampère- x sees Ω_A , Faraday sees Ω_F . Their product, not either alone, is what enters the local dispersion relation (§5); the split originates in the model keeping $A_x = 0$ fixed while $E_x \neq 0$, and is inherited from the two distinct covariant structures in `khaxn.pdf` (cf. its eqs. (2)/(3) vs (16)/(19)).

Step 4.2.3 (Ampère- z). Linearize (3.6): no commutator (§3.1), curl term $\partial_x B_y \rightarrow b'$ (backward difference in the code; see §6.3). The current perturbation now survives at first order because the beams carry background $v_z = \pm v$:

$$\delta J_z^{2,3} = -\left[n_A \delta Q_A^{2,3} v_{zA} + n_B \delta Q_B^{2,3} v_{zB}\right] = -v(\delta Q_A^{2,3} - \delta Q_B^{2,3}), \quad (4.7)$$

using $n_A = n_B = 1$, $v_{zA} = -v_{zB} = v$ (density and velocity perturbations multiply the *zero* background $Q^{2,3}$, hence drop). In circular form $\delta J_z^c = -v(q_A - q_B)$ and

$$\gamma e_z = b' + v(q_A - q_B). \quad (4.8)$$

Step 4.2.4 (Potential). Directly from (3.8):

$$\gamma a = -e_z. \quad (4.9)$$

Step 4.2.5 (Precession, beam A). Take (3.4) in advective form for beam A ($Q_A^1 = 1$ background, $v_{zA} = v$). The advection term contributes the Doppler shift $ik_z v q_A$; the source needs the 2- and 3-components of $\vec{Q} \times \vec{A}_z$ keeping first-order terms. With $\vec{Q} = (1, Q^2, Q^3)$ and $\vec{A}_z = (A_{z1}, a_2, a_3)$:

$$(\vec{Q} \times \vec{A}_z)^2 = Q^3 A_{z1} - Q^1 a_3 = Q^3 A_{z1} - a_3, \quad (4.10)$$

$$(\vec{Q} \times \vec{A}_z)^3 = Q^1 a_2 - Q^2 A_{z1} = a_2 - Q^2 A_{z1}. \quad (4.11)$$

Combine as (2) + i(3):

$$(\vec{Q} \times \vec{A}_z)^c = Q^3 A_{z1} - a_3 + ia_2 - iQ^2 A_{z1} = i(a_2 + ia_3) - iA_{z1}(Q^2 + iQ^3) = ia - iA_{z1} q_A. \quad (4.12)$$

Hence $\gamma q_A + ik_z v q_A = \alpha v(ia - iA_{z1} q_A)$, i.e.

$$\gamma q_A = i\alpha v a - iv(k_z + \alpha A_{z1}) q_A = i\alpha v a - iv\Omega_A q_A. \quad (4.13)$$

The combination is meaningful: the beam's Doppler shift $k_z v$ and its precession frequency $\alpha v A_{z1}$ enter only through the single product $v\Omega_A$.

Step 4.2.6 (Precession, beam B). Repeat with $Q_B^1 = -1$, $v_{zB} = -v$:

$$(\vec{Q} \times \vec{A}_z)^2 = Q^3 A_{z1} + a_3, \quad (\vec{Q} \times \vec{A}_z)^3 = -a_2 - Q^2 A_{z1}, \quad (4.14)$$

$$(\vec{Q} \times \vec{A}_z)^c = Q^3 A_{z1} + a_3 - ia_2 - iQ^2 A_{z1} = -ia - iA_{z1} q_B. \quad (4.15)$$

Then $\gamma q_B - ik_z v q_B = \alpha(-v)(-ia - iA_{z1} q_B)$, i.e.

$$\gamma q_B = i\alpha v a + iv\Omega_A q_B. \quad (4.16)$$

The two beams are driven identically ($+i\alpha v a$) but Doppler-rotate in opposite senses ($\mp iv\Omega_A$): both charge-sign flips (in the source and in the Doppler term) are compensated by the velocity sign flip except in the rotation term.

Collecting (4.5)–(4.16), the eigenvalue system $\gamma\Psi = L\Psi$, $\Psi = (b, e_x, e_z, a, q_A, q_B)$:

$$\boxed{\begin{aligned} \gamma b &= -i\Omega_F e_x + e'_z & \gamma a &= -e_z \\ \gamma e_x &= -i\Omega_A b & \gamma q_A &= i\alpha v a - iv\Omega_A q_A \\ \gamma e_z &= b' + v(q_A - q_B) & \gamma q_B &= i\alpha v a + iv\Omega_A q_B \end{aligned}} \quad (4.17)$$

4.3 Helicity conjugation and parity

The conjugate pairs $\bar{f} = f^2 - if^3$ satisfy (4.17) with $A_{z1} \rightarrow -A_{z1}$ (equivalently $u \rightarrow -u$, $\Omega_A \leftrightarrow \Omega_F$), because (4.3) acquires the opposite sign of iA_{z1} under conjugation. Separately, the background has the parity $v(-\xi) = -v(\xi)$, $A_{z1}(-\xi) = A_{z1}(\xi)$. Combining the two operations $\xi \rightarrow -\xi$ and helicity conjugation maps (4.17) onto itself with the same γ : *every eigenmode of one circular polarization localized on one side of the layer has a mirror partner of the opposite polarization on the other side, at exactly the same growth rate.* This is the symmetry behind the one-sided eigenfunctions found throughout the program (§6.2), and behind the helicity bookkeeping of the extraction pipeline (Appendix A).

4.4 Correspondence with the given frequency-domain theory; scope

Eliminating e_x, e_z, q_A, q_B, b from (4.17) (§6) produces beam response denominators $\gamma \pm iv\Omega_A$. Under $\omega = i\gamma$,

$$\gamma \pm iv\Omega_A = -i(\omega \mp v\Omega_A) = -i[(\omega \mp k_z v) \mp \alpha v A_{z1}], \quad (4.18)$$

which is exactly the structure $\Omega + \alpha\Phi_0^1$ of the given eq. (1.1) ($\Omega = \omega - k_z V_{0z}$ the Doppler-shifted frequency, $\Phi_0^1 = V_{0z} A_{0z}^1$ the effective background coupling), one factor per beam. Likewise the left-hand operator of (1.1) contains $k_z^2 - \omega^2 - \alpha^2 A_{0z}^2 = \gamma^2 + k_z^2 - u^2 / \dots$, the factor $\mathcal{D}$ of §6. The time-domain model therefore linearizes into precisely the given theory, with two deliberate scope differences:

1. no background scalar potential / background electric field ($\phi_0^1 = E_{0x}^1 = 0$ in the simulation; the $\alpha^2(\phi_0^1)^2$ term of (1.1) is absent), and
2. no background B_{y1} coupling in Ampère- x and no fluid density/momentum perturbations in the state vector (their combined effect on the shear branch is bounded at the few-percent level by the simulation-vs-solver agreement; the $a_y^\pm$ sector of the given theory decouples at this order and is not simulated).

5 Local dispersion: the tachyonic outer branch and the local drive branch

Here we freeze the background locally ($A_{z1} = A$, v constant) and take plane waves in x ($\partial_x \rightarrow ik_x$). The system (4.17) becomes a 6×6 matrix problem; at $k_x = 0$ it factorizes into two independent blocks whose exact dispersion relations we now derive.

5.1 Block factorization at $k_x = 0$

Setting $e'_z \rightarrow ik_x e_z = 0$ and $b' \rightarrow ik_x b = 0$ in (4.17): the pair (b, e_x) closes on itself,

$$\gamma b = -i\Omega_F e_x, \quad \gamma e_x = -i\Omega_A b \quad (\text{EM block}), \quad (5.1)$$

and the quadruple (e_z, a, q_A, q_B) closes on itself,

$$\gamma e_z = v(q_A - q_B), \quad \gamma a = -e_z, \quad \gamma q_{A,B} = i\alpha v a \mp iv\Omega_A q_{A,B} \quad (\text{charge block}). \quad (5.2)$$

5.2 EM block: the tachyonic branch

Step 5.2.1. Multiply the two equations of (5.1):

$$\gamma^2 b e_x = (-i\Omega_F e_x)(-i\Omega_A b) = -\Omega_A \Omega_F b e_x \implies \gamma^2 = -\Omega_A \Omega_F = \alpha^2 A^2 - k_z^2. \quad (5.3)$$

Step 5.2.2 (Interpretation). The two circular polarizations carry adjoint color charge ± 1 under the background color-1 rotation; the covariant z -derivative $D_z = \partial_z \mp i\alpha A_{z1}$ gives them effective wavenumbers $k_z \pm \alpha A$. The product $\Omega_A \Omega_F = (k_z - u)(k_z + u) = k_z^2 - u^2$ is negative wherever the background shift exceeds the wavenumber, and there one polarization becomes *tachyonic*: $\gamma^2 > 0$. This is the charged-vector-on-strong-color-background instability of the Nielsen–Olesen class (Nielsen & Olesen 1978), realized here in 2D with A playing the role of the strong background field.

Step 5.2.3 (Instability region and local envelope). With the production background (1.8), $u(\xi) = \alpha V_0 \ln \cosh \xi$ and (1.9), the local growth rate

$$\gamma_{\text{loc}}(\xi) = \sqrt{u(\xi)^2 - k_z^2} \simeq \sqrt{(\alpha V_0)^2 (|\xi| - \ln 2)^2 - k_z^2} \quad (5.4)$$

is real exactly for

$$|\xi| > \xi_{\text{crit}} \simeq \frac{k_z}{\alpha V_0} + \ln 2 \quad (5.5)$$

and grows asymptotically linearly, $\gamma_{\text{loc}} \rightarrow \alpha V_0 |\xi|$.

Step 5.2.4 (Finite k_x only subtracts). Keep $k_x \neq 0$ but drop the charge coupling (outer region, v at its plateau; the charge terms decouple from the growing branch — verified numerically at all outer eigenmode peaks). Then (4.17) retains e_z : from $\gamma e_z = ik_x b$, $\gamma b = -i\Omega_F e_x + ik_x e_z$, $\gamma e_x = -i\Omega_A b$:

$$\gamma^2 b = -i\Omega_F (\gamma e_x) + ik_x (\gamma e_z) = -\Omega_A \Omega_F b - k_x^2 b \implies \gamma^2 = u^2 - k_z^2 - k_x^2. \quad (5.6)$$

$\gamma_{\text{loc}}(\xi)$ of (5.4) is therefore the *pointwise envelope* over all x -structures; global modes must oscillate ($k_x^2 > 0$ costs growth) and sit below it.

Remark 5.1 (Energy source and the freeze artifact). By (3.20), this branch is powered entirely by the frozen A_{z1} ; the model as posed makes it a true linear instability with a bottomless supply. Its runaway character (not its existence) is the freeze artifact; windows (sponge/cut) placed inside ξ_{crit} are the legitimate excision of this branch from shear-physics measurements. The self-consistent test (§13.2) shows the unfrozen back-reaction does not deplete the background either — it deepens it locally and stabilizes only weakly and globally.

Corollary 5.2 (High- α window collapse). *A window of radius ξ_w excludes the tachyonic branch only if $\xi_w < \xi_{\text{crit}} = k_z/(\alpha V_0) + \ln 2$. As α grows at fixed k_z, V_0 , $\xi_{\text{crit}} \rightarrow \ln 2$: eventually no window can both contain the shear-branch well (which needs ξ_w of order several) and exclude the tachyon. This is the mechanism of the early tachyonic blow-ups in the $\alpha \geq 6$ corner of the EPS-scan $v2$ campaign: at $\alpha = 6$, $V_0 = 0.05$, $k_z = 2$, $\xi_{\text{crit}} = 7.4$, already inside the vetted windows, and the failure maps to α , not k_z , exactly as (5.5) predicts.*

Remark 5.3 (The “precession cascade” scale $\gamma_{\text{casc}} \approx \alpha V_0$). Near the layer ($|\xi| = \mathcal{O}(1)$, $|A_{z1}| \lesssim V_0$), the same color-mixing terms rotate the color-2/3 sector at the local rate $\alpha|A_{z1}| \sim \alpha V_0$. Seeding campaigns observed the resulting build-up of $A_{z1}^{2,3}$ at $\gamma_{\text{casc}} \approx \alpha V_0$ (0.20–0.24 at $\alpha = 2$, $V_0 = 0.1$), which can mask slower KH growth — the motivation for eigenfunction seeding (§15). It is the same mechanism as (5.3) evaluated at the layer-scale $|A_{z1}|$ rather than the outer plateau.

5.3 Charge block: the local drive branch

Step 5.3.1 (Solve the two precession equations). From (5.2),

$$q_A = \frac{i\alpha v a}{\gamma + i v \Omega_A}, \quad q_B = \frac{i\alpha v a}{\gamma - i v \Omega_A}. \quad (5.7)$$

Step 5.3.2 (Beam-difference response).

$$q_A - q_B = i\alpha v a \left[\frac{1}{\gamma + i v \Omega_A} - \frac{1}{\gamma - i v \Omega_A} \right] = i\alpha v a \frac{-2i v \Omega_A}{\gamma^2 + v^2 \Omega_A^2} = \frac{2\alpha v^2 \Omega_A}{\mathcal{R}} a, \quad \mathcal{R} \equiv \gamma^2 + v^2 \Omega_A^2. \quad (5.8)$$

Step 5.3.3 (Close through e_z and a). $\gamma e_z = v(q_A - q_B) = \frac{2\alpha v^3 \Omega_A}{\mathcal{R}} a \equiv g a$ and $e_z = -\gamma a$ give

$$-\gamma^2 a = g a \quad \Rightarrow \quad \boxed{\gamma^2 = -g, \quad g(\xi; \gamma) \equiv \frac{2\alpha v^3 \Omega_A}{\gamma^2 + v^2 \Omega_A^2}.} \quad (5.9)$$

On the side of the layer where $v^3 \Omega_A < 0$, $g < 0$ and the block grows: this is the *local (homogeneous) limit of the shear-branch drive*. The function g defined here is exactly the drive that appears in the reduced ODE (§6); equation (5.9) is its $k_x \rightarrow 0$ limit, and its maximization over the background will produce the growth ceiling in §7.

5.4 Airy estimate for wall-bounded outer modes

Global tachyonic modes live in the annulus between ξ_{crit} and the window radius ξ_w (hard cut: Dirichlet wall). Their rates follow from a one-dimensional bound-state problem in the local potential $U(x) \equiv \gamma_{\text{loc}}^2(x)$, increasing outward:

$$a'' + (U(x) - \gamma^2) a = 0, \quad (5.10)$$

with $a = 0$ at the wall x_w and decay toward the layer.

Step 5.4.1 (Linearize at the wall). Near x_w write $U(x) = U_w - U'(x_w - x)$ with $U_w = U(x_w)$, $U' = dU/dx > 0$. In the inward coordinate $t = x_w - x \geq 0$:

$$\frac{d^2 a}{dt^2} - (\gamma^2 - U_w + U' t) a = 0. \quad (5.11)$$

Step 5.4.2 (Airy form). Substitute $\tau = (U')^{1/3} t + \tau_0$ with $\tau_0 = (\gamma^2 - U_w)/(U')^{2/3}$:

$$\frac{d^2 a}{d\tau^2} = \tau a, \quad (5.12)$$

whose solution decaying for $\tau \rightarrow +\infty$ (deep inside, where $U < \gamma^2$) is $\text{Ai}(\tau)$.

Step 5.4.3 (Wall condition). The Dirichlet wall at $t = 0$ requires $\text{Ai}(\tau_0) = 0$. The least negative zero of Ai is $-a_1 = -2.338\dots$, so the deepest (fastest) mode has $\tau_0 = -a_1$:

$$\boxed{\gamma^2 \simeq U_w - 2.338 (U')^{2/3}} \quad (5.13)$$

Corollary 5.4 (Why tight windows kill the branch outright). *For a tight window, $U_w = \gamma_{\text{loc}}^2(\xi_w)$ is small (the wall sits just past ξ_{crit}) while the slope $U' \simeq 2(\alpha V_0)^2(\xi_w - \ln 2)/\varepsilon_s$ is not; (5.13) then goes negative: the annulus holds no bound state at all, and windowing does not merely reduce the outer branch — it removes it. Measured global rates at the widest tested annuli reach $0.78\times$ the envelope with a discrete ladder below, in line with the quantization cost in (5.13).*

6 Exact scalar reduction of the six-field system

The six-field system (4.17) contains only one x -derivative chain ($e'_z \rightarrow b \rightarrow b'$); everything else is algebraic. It therefore reduces *exactly* to one second-order ODE for the potential amplitude $a(x)$. This reduction, verified to machine precision against the ARPACK eigenvalues of the discrete six-field operator, is the backbone of the exact-action theory.

6.1 Continuum elimination

Step 6.1.1 (Eliminate e_x). From (4.5): $e_x = -i\Omega_A b/\gamma$.

Step 6.1.2 (Eliminate e_z). From (4.9): $e_z = -\gamma a$.

Step 6.1.3 (Express b through a'). Insert both into Faraday (4.6):

$$\gamma b = -i\Omega_F \left(-\frac{i\Omega_A}{\gamma} b \right) + (-\gamma a)' = -\frac{\Omega_A \Omega_F}{\gamma} b - \gamma a'. \quad (6.1)$$

Multiply by γ and collect b :

$$b(\gamma^2 + \Omega_A \Omega_F) = -\gamma^2 a' \quad \implies \quad b = -\frac{\gamma^2 a'}{\mathcal{D}}, \quad \boxed{\mathcal{D}(x) \equiv \gamma^2 + \Omega_A \Omega_F = \gamma^2 + k_z^2 - u(x)^2} \quad (6.2)$$

$\mathcal{D}$ is the electromagnetic factor: $\mathcal{D} < 0$ is exactly the tachyonic region of §5.2 (one polarization sees a negative effective k^2); the shear branch lives where $\mathcal{D} > 0$.

Step 6.1.4 (Charge response). Equations (5.7)–(5.8) hold pointwise with $v = v(\xi)$, $\Omega_A = \Omega_A(\xi)$:

$$q_A - q_B = \frac{2\alpha v^2 \Omega_A}{\mathcal{R}} a, \quad \mathcal{R}(x) = \gamma^2 + v^2 \Omega_A^2. \quad (6.3)$$

Step 6.1.5 (Assemble Ampère- z). Insert $e_z = -\gamma a$, (6.2) and the charge response into (4.8):

$$-\gamma^2 a = \left(-\frac{\gamma^2 a'}{\mathcal{D}} \right)' + \frac{2\alpha v^3 \Omega_A}{\mathcal{R}} a \quad (6.4)$$

and rearrange:

$$\boxed{\left(\frac{\gamma^2 a'}{\mathcal{D}} \right)' = (\gamma^2 + g) a, \quad g(x) = \frac{2\alpha v^3 \Omega_A}{\gamma^2 + v^2 \Omega_A^2}} \quad (6.5)$$

— one Sturm–Liouville-type equation, nonlinear in the eigenvalue γ (which appears in $\mathcal{D}$, g , and the right side), but scalar.

6.2 Anatomy of the reduced equation

- **$\mathcal{D}$ (EM factor).** Sign changes of $\mathcal{D}$ separate the propagating/trapped shear-branch region from the tachyonic outer region: one mechanism — the covariant shift u — creates both the confining well and, past $|\xi| = \xi_{\text{crit}}$, the runaway. (“The well and the tachyon are two sides of one term.”)
- **g (drive).** The *only* drive of the shear branch, and precisely the local-drive function of (5.9). Its three powers of v have distinct origins: one from the precession source ($i\alpha v a$ in (4.13)), one from the beam Doppler asymmetry (the $v\Omega_A/\mathcal{R}$ structure of (5.8)), one from the current ($\delta J_z = -v(q_A - q_B)$). Counter-streaming enters only through the even combination $\mathcal{R} = \gamma^2 + v^2 \Omega_A^2$.
- **One-sidedness.** $g \propto v^3$ is odd in ξ (for $\Omega_A > 0$): each circular polarization is driven on one side of the layer only; the mirror image is carried by the conjugate polarization at the same $|\gamma|$ (§4.3). This explains the one-sided eigenfunctions seen in all solver outputs.

- **WKB form.** Writing $a \propto \exp(\pm i \int \sqrt{Q} dx)$ brings (6.5) to the standard form with the exact local wavenumber

$$Q(x; \gamma) = - \frac{\mathcal{D}(\gamma^2 + g)}{\gamma^2} \quad (6.6)$$

(the flux form contributes the $\mathcal{D}$ weight; a well requires $Q > 0$, i.e. $\mathcal{D} > 0$ together with $g < -\gamma^2$).

6.3 Exactness at the discrete level; the two wall faces

The eigensolver uses a forward difference for $\partial_x e_z$ (Faraday row) and a backward difference for $\partial_x b$ (Ampère- z row), matching the simulation's Yee stagger. Repeat the elimination on the lattice ($x_i = i dx$):

Step 6.3.1. The discrete Faraday/Ampère- x pair gives, per node i ,

$$b_i \frac{\mathcal{D}_i}{\gamma} = \frac{e_{z,i+1} - e_{z,i}}{dx} \implies b_i = - \frac{\gamma^2}{\mathcal{D}_i} \frac{a_{i+1} - a_i}{dx}, \quad (6.7)$$

i.e. b_i lives on the *face* between nodes i and $i+1$, with $\mathcal{D}$ evaluated at the left node of the face.

Step 6.3.2. The discrete Ampère- z row then reads

$$-\gamma^2 a_i = \frac{b_i - b_{i-1}}{dx} + g_i a_i = - \frac{\gamma^2}{dx^2} \left[\frac{a_{i+1} - a_i}{\mathcal{D}_i} - \frac{a_i - a_{i-1}}{\mathcal{D}_{i-1}} \right] + g_i a_i, \quad (6.8)$$

the exact flux-form finite difference of (6.5). No approximation was made: the $6N \times 6N$ eigenproblem and the $N \times N$ nonlinear problem (6.8) have identical spectra on the shared modes. Verified: Newton iteration on (6.8) reproduces the six-field ARPACK eigenvalues to relative difference 0.0 (below 10^{-10}) at seven points spanning $\alpha = 1-3$, $V_0 = 0.03-0.1$, $k_z = 1-7$.

Step 6.3.3 (Wall faces). A hard cut (`xi_cut`) zeroes all fields outside radius ξ_w . In the reduction the two sides of the wall behave differently:

- where the *right* neighbor is outside, $e_{z,i+1}$ is a Dirichlet ghost ($e_z = 0$, hence $a = 0$): the face flux uses a zero value — a Dirichlet face;
- where the *left* neighbor is outside, b_{i-1} is dropped: the face flux itself vanishes — a zero-flux (Neumann, $b = 0 \equiv a' = 0$) face.

The mode side of the wall is therefore Neumann for a , the far side Dirichlet; the two faces are *not* equivalent, and the Neumann condition on the mode side is what enters the quantization phase in §8.

7 The drive: precession resonance, the growth ceiling, and natural units

7.1 The saturated drive $G(w)$

In the saturated region $|\xi| \gtrsim 2$: $v = \pm V_0$ and $u = \alpha V_0(|\xi| - \ln 2)$ by (1.9). On the driven side ($v^3 \Omega_A < 0$; concretely $\xi < 0$ for the $2 + i3$ polarization when $\Omega_A > 0$) the magnitude of the drive is

$$-g = G(w) \equiv \frac{2\alpha V_0^3 w}{\gamma^2 + V_0^2 w^2}, \quad w \equiv \Omega_A = k_z - u(\xi) \quad (7.1)$$

— a function of position only through the local covariant wavenumber w , which decreases linearly outward from k_z at the layer.

7.2 Resonance and the exact ceiling

Step 7.2.1 (Maximize G over w).

$$\frac{dG}{dw} = 2\alpha V_0^3 \frac{(\gamma^2 + V_0^2 w^2) - w(2V_0^2 w)}{(\gamma^2 + V_0^2 w^2)^2} = 2\alpha V_0^3 \frac{\gamma^2 - V_0^2 w^2}{(\gamma^2 + V_0^2 w^2)^2} = 0 \implies w^* = \frac{\gamma}{V_0}. \quad (7.2)$$

The condition $V_0 w^* = \gamma$ says: the mode grows fastest where the *beam-frame precession frequency* $v\Omega_A = V_0 w$ matches the growth rate — a precession resonance surface in the flow. The peak value is

$$G_{\max} = G(w^*) = \frac{2\alpha V_0^3 (\gamma/V_0)}{2\gamma^2} = \frac{\alpha V_0^2}{\gamma}. \quad (7.3)$$

Step 7.2.2 (Ceiling, derivation 1 (well existence)). By (6.6), a confining well ($Q > 0$ with $\mathcal{D} > 0$) requires $g < -\gamma^2$, i.e. $G > \gamma^2$ somewhere. Since $G \leq G_{\max}$,

$$\gamma^2 \leq \frac{\alpha V_0^2}{\gamma} \iff \boxed{\gamma^3 \leq \alpha V_0^2} \quad (7.4)$$

independent of k_z , ε_s , and window geometry.

Step 7.2.3 (Ceiling, derivation 2 (local drive branch)). Alternatively, the homogeneous charge-block branch (5.9) gives $\gamma^2 = G(w)$ pointwise; maximizing the implicit solution over w means solving $\gamma^2 = G(w^*) = \alpha V_0^2/\gamma$, i.e. $\gamma = (\alpha V_0^2)^{1/3}$ exactly. The ceiling is thus the maximum local growth rate of the drive branch, attained on the resonance surface; any global (quantized, windowed) mode sits below it. Measured dominant-branch peaks reach 95–99% of $(\alpha V_0^2)^{1/3}$ in every validated series.

7.3 Natural units and the universal well

Define, as in (1.11), $\Gamma = (\alpha V_0^2)^{1/3}$, $K = (\alpha/V_0)^{1/3}$, and hatted variables

$$\gamma = \Gamma \hat{\gamma}, \quad k_z = K \hat{k}, \quad w = K \hat{w}, \quad u = K \hat{u}. \quad (7.5)$$

Step 7.3.1 (Scale out the drive). Using $V_0^2 K^2 = V_0^2 (\alpha/V_0)^{2/3} = (\alpha V_0^2)^{2/3} = \Gamma^2$ and $\alpha V_0^3 K = \alpha^{4/3} V_0^{8/3} = \Gamma^4$:

$$G(w) = \frac{2\alpha V_0^3 K \hat{w}}{\Gamma^2 \hat{\gamma}^2 + V_0^2 K^2 \hat{w}^2} = \Gamma^2 \frac{2\hat{w}}{\hat{\gamma}^2 + \hat{w}^2} \equiv \Gamma^2 \hat{G}(\hat{w}). \quad (7.6)$$

The drive is universal in hatted variables: no α , V_0 , or ε_s remains in $\hat{G}$.

Step 7.3.2 (Universal well condition and edges). $G > \gamma^2$ becomes $\hat{G} > \hat{\gamma}^2$:

$$\boxed{\hat{\gamma}^2 (\hat{\gamma}^2 + \hat{w}^2) < 2\hat{w}} \quad (7.7)$$

The edges solve $\hat{\gamma}^2 \hat{w}^2 - 2\hat{w} + \hat{\gamma}^4 = 0$:

$$\hat{w}_{\pm} = \frac{1 \pm \sqrt{1 - \hat{\gamma}^6}}{\hat{\gamma}^2}, \quad (7.8)$$

real only for $\hat{\gamma} \leq 1$ — a third, independent derivation of the ceiling (7.4). As $\hat{\gamma} \rightarrow 1$ the well closes onto the resonance point $\hat{w} = 1$ (i.e. $w = K$, $\gamma/V_0 = K$).

Step 7.3.3 (Provable non-collapse in αV_0). Rates scale as $(\alpha V_0^2)^{1/3}$ and wavenumbers as $(\alpha/V_0)^{1/3}$: no combination of the two is a function of the product αV_0 alone. This proved that the survey-era attempt to collapse the data on αV_0 had to fail, and supplied the variables for the successful collapse test (T2.5: $\gamma_{\text{peak}}/\Gamma = 0.977 \pm 0.011$ across $\alpha = 1\text{--}3$, $V_0 = 0.03\text{--}0.1$ at fixed window group and $\varepsilon_s = 0.15$).

7.4 The resonance surface in space

Setting $w(\xi) = k_z - u(\xi) = w^* = \gamma/V_0$ and inverting (1.9):

$$\xi_{\text{res}} \simeq \frac{k_z - \gamma/V_0}{\alpha V_0} + \ln 2 \quad (7.9)$$

— the resonance surface moves *outward linearly with k_z* . The eigenmode peak tracks it (measured: peak ξ from -1 to -16 across $k_z = 0.5$ – 10 at the C25 parameters, -36 at cut 40), which is the geometric fact behind the confinement-set dispersion peak of §9.

8 Quantization of the shear branch: models A and B

The well is shallow: the branch holds *less than a quarter wavelength* of action at production parameters (measured $S = 0.31\pi$ – 0.46π), so naive Bohr–Sommerfeld $(n + \frac{1}{2})\pi$ quantization fails and the connection conditions at the two edges do all the work.

8.1 Geometry of the well

Fix the driven side ($\xi < 0$ for the $2 + i3$ pair). Moving inward from the wall:

- **Outer edge:** the window radius ξ_w (hard cut $\rightarrow$ zero-flux/Neumann face for a , §6.3); or, if the window is loose, the free turning point where G falls back to γ^2 (at $w = \hat{w}_-$ side, eq. (7.8)).
- **Inner edge:** the “knee” ξ_0 where the v^3 factor switches the drive off over $\lesssim 1$ ξ -unit. Since $g = -G |\tanh \xi|^3$ near the layer-side edge of the saturated region, the well ends where

$$G |\tanh \xi_0|^3 = \gamma^2 \quad \implies \quad \tanh^3 \xi_0 = \frac{\gamma^2}{G}, \quad (8.1)$$

giving $\xi_0 \approx 1.4$ – 2 at production parameters (the code’s model A uses the fixed representative $\xi_0 = 1.5$).

- **Interior:** $Q > 0$ from (6.6). In the window-limited regime relevant near the dispersion peak, $u \ll k_z$ across the well, so $\mathcal{D} \approx \gamma^2 + k_z^2$ and $G \approx G(k_z)$ vary by $< 5\%$ across the well — the numerically confirmed *square-well regime*.
- **Beyond the inner edge** ($|\xi| < \xi_0$): $g \rightarrow 0$, so $Q \rightarrow -\mathcal{D} \approx -(\gamma^2 + k_z^2)$: evanescent with decay constant

$$\kappa = \sqrt{\gamma^2 + k_z^2} \approx k_z. \quad (8.2)$$

8.2 The sharp-edge connection formula

Step 8.2.1 (Interior solution). Let $k_x = \sqrt{Q}$ (constant in the square-well regime) and measure distance s inward from the outer wall. The Neumann face ($a' = 0$ at $s = 0$) selects

$$a(s) = \cos(k_x s). \quad (8.3)$$

Step 8.2.2 (Log-derivative at the inner edge). At the inner edge $s = \Delta$ (well width), with total action $S = k_x \Delta$:

$$\left. \frac{a'}{a} \right|_{s=\Delta^-} = -k_x \tan S. \quad (8.4)$$

Step 8.2.3 (Evanescent side). For $s > \Delta$ the solution decaying toward the layer is $a \propto e^{-\kappa(s-\Delta)}$, with $a'/a = -\kappa$. The drive switches off over a distance short compared with $2\pi/k_x$ (the “sharp step”), so a and a' are continuous across it:

$$-k_x \tan S = -\kappa \quad \implies \quad \boxed{\tan S = \frac{\kappa}{k_{\text{edge}}}} \quad (k_{\text{edge}} = k_x \text{ at the inner edge}), \quad (8.5)$$

with $S \in (0, \pi/2)$ for the nodeless $n = 0$ mode and $S \in (n\pi, n\pi + \pi/2)$ for the n -th overtone. Because $\tan S > 0$ requires $S < \pi/2 \pmod{\pi}$, the branch is *quantization-marginal*: it holds less than a quarter wave, and γ adjusts to sit close below the ceiling so that the well is exactly this shallow.

8.3 Model A: closed-form dispersion relation

Model A adopts the square-well regime literally and closes everything analytically.

Step 8.3.1 (Assumptions). (i) $u \ll k_z$ over the well, so $w \approx k_z$ and the drive is the *Doppler-detuned* value

$$G = G(k_z) = \frac{2\alpha V_0^3 k_z}{\gamma^2 + V_0^2 k_z^2}; \quad (8.6)$$

(ii) $\mathcal{D} \approx \gamma^2 + k_z^2$ constant; (iii) well width $\Delta x = \varepsilon_s (\xi_w - \xi_0)$ with $\xi_0 = 1.5$.

Step 8.3.2 (Interior wavenumber). From (6.6) with these approximations,

$$k_x = \sqrt{Q} = \sqrt{\frac{(\gamma^2 + k_z^2)(G - \gamma^2)}{\gamma^2}} = \sqrt{\gamma^2 + k_z^2} \frac{\sqrt{G - \gamma^2}}{\gamma} \approx k_z X, \quad \boxed{X \equiv \frac{\sqrt{G - \gamma^2}}{\gamma}} \quad (8.7)$$

(using $k_z \gg \gamma$ in the prefactor, consistent with (i)).

Step 8.3.3 (Quantization closes on X alone). Insert $k_x = k_z X$ and $\kappa \approx k_z$ into (8.5):

$$S = k_x \Delta x = \underbrace{\varepsilon_s (\xi_w - \xi_0) k_z}_{\equiv L} X, \quad \tan S = \frac{k_z}{k_z X} = \frac{1}{X} \quad \implies \quad \boxed{L X = \arctan \frac{1}{X}} \quad (8.8)$$

— a single transcendental equation for X that contains *no* γ : the geometry factor L (window, shear width, wavenumber) fixes X by itself. X decreases monotonically as L grows (looser window $\Rightarrow$ smaller interior wavenumber needed).

Step 8.3.4 (Recover γ). By the definition of X , $G = \gamma^2(1 + X^2)$; equate to (8.6):

$$\gamma^2 (1 + X^2) (\gamma^2 + V_0^2 k_z^2) = 2\alpha V_0^3 k_z. \quad (8.9)$$

This is a quadratic in γ^2 :

$$\gamma^4 + V_0^2 k_z^2 \gamma^2 - \frac{2\alpha V_0^3 k_z}{1 + X^2} = 0, \quad (8.10)$$

whose positive root is

$$\boxed{\gamma^2 = \frac{1}{2} \left[-V_0^2 k_z^2 + \sqrt{V_0^4 k_z^4 + \frac{8\alpha V_0^3 k_z}{1 + X^2}} \right]} \quad (8.11)$$

— model A. Validated within $\sim 3\%$ of the σ -chased dominant eigenvalues near and above the dispersion peak, and it identifies the peak within ± 0.5 in k_z .

8.4 Model B: exact-action quantization

Model B keeps the exact $Q(x; \gamma)$ of (6.6) — no square-well or $w \approx k_z$ approximation — and root-finds the quantization phase. For trial γ :

1. compute Q on a fine grid of the driven side; find the largest contiguous region $Q > 0$ (the well) inside the window radius ξ_w ;
2. exact action $S(\gamma) = \int_{\text{well}} \sqrt{Q} dx$;
3. edge wavenumber $k_{\text{edge}} = \sqrt{Q}$ evaluated just inside the *inner* edge; $\kappa = \sqrt{\gamma^2 + k_z^2}$;
4. connection phase Φ :

$$\Phi = \begin{cases} \arctan(\kappa/k_{\text{edge}}) & \text{well attached to the tanh}^3 \text{ knee (sharp inner step),} \\ \pi/4 & \text{well detached (soft inner turning point),} \end{cases} \quad (8.12)$$

with an additional $+\pi/4$ if the outer edge is a free turning point rather than the wall, plus $n\pi$ for overtones. The $\pi/4$ terms are the standard Airy connection phases at first-order turning points (the linear-turning-point matched asymptotics that replace the sharp-step matching of §8.2); the Neumann wall contributes phase 0 (cosine antinode at the wall, exactly as in (8.5)).

5. solve $S(\gamma) = \Phi(\gamma)$, scanning γ downward from just below the ceiling so the first bracket found is $n = 0$.

Validation: across 12 series spanning $\alpha = 1\text{--}5$, $V_0 = 0.03\text{--}0.2$, windows $\xi_w = 5\text{--}40$, $k_z = 0.5\text{--}10$, model B matches the σ -chased dominant ($n = 0$) six-field eigenvalues to $\leq 2\%$ (median $\leq 0.2\%$) at every $k_z \geq 1.5$; the worst deviations (6–9%) occur at $k_z \lesssim 1$ where the well is widest and least square.

8.5 Overtone ladder and eigenvalue-selection hygiene

With $S(\gamma)$ decreasing in γ , the roots of $S(\gamma) = \Phi + n\pi$ form a ladder $\gamma_0 > \gamma_1 > \dots$ with spacing

$$\Delta\gamma \simeq \frac{\pi}{|\partial S/\partial \gamma|}, \quad (8.13)$$

a few $\times 10^{-2} \text{ TU}^{-1}$ at loose windows. A shift-invert eigensolver returns the eigenvalue nearest its target σ ; the legacy default $\sigma = 0.55 \gamma_{\text{WKB}}(\text{quartic})$ lands mid-ladder wherever the quartic underestimates (low α , or k_z above the quartic’s peak), silently selecting an overtone. Consequence (referee-critical, discovered via this theory): at $(\alpha, V_0, k_z) = (1, 0.05, 4)$, window 20, the dominant mode has $\gamma = 0.134$ but default- σ returned 0.106 and the cached survey curve carried 0.082 — an $n \approx 3$ overtone; the eigenmode-seeded simulation then faithfully grew the overtone it was seeded with (plateau-audited 0.081). *Simulation-vs-solver agreement certifies numerics, not dominant-mode selection.* All reference data now use σ -chasing (re-shifting σ upward until the found window is clear).

9 The dispersion curve: confinement-set peak and windowed roll-off

9.1 No intrinsic k_z maximum

In an unbounded domain nothing stops the eigenmode from following the resonance surface (7.9) outward as k_z grows: the drive at resonance is always $G_{\text{max}} = \alpha V_0^2/\gamma$, independent of k_z , so $\gamma(k_z)$

risers monotonically and saturates at the ceiling (7.4) while the mode migrates outward. Directly demonstrated (eigensolver, $\alpha = 2$, $V_0 = 0.05$, cut 40): γ climbs to 0.9955Γ by $k_z = 10$ with no turnover, mode peak at $\xi = -36$. The non-monotonic measured dispersion is therefore *windowed physics*, not an intrinsic wavelength selection.

9.2 The peak: resonance surface meets the window

Step 9.2.1. For k_z below the peak, the resonance surface $\xi_{\text{res}}(k_z)$ of (7.9) lies inside the window and the mode enjoys the full resonant drive: γ near the ceiling.

Step 9.2.2. Once ξ_{res} reaches the window radius ξ_w , the well is truncated at the wall value of the covariant wavenumber, $u_w = \alpha V_0(\xi_w - \ln 2)$, the resonance is no longer accessible, and only the detuned drive $G(k_z)$ of (8.6) remains: γ rolls off (next subsection). The peak is the crossover $\xi_{\text{res}}(k_z) = \xi_w$:

$$\frac{k_z - \gamma/V_0}{\alpha V_0} + \ln 2 = \xi_w. \quad (9.1)$$

Step 9.2.3. At the crossover the mode still runs at essentially the ceiling, so substitute $\gamma = \Gamma = (\alpha V_0^2)^{1/3}$, i.e. $\gamma/V_0 = (\alpha/V_0)^{1/3} = K$:

$$k_z^{\text{peak}} \simeq \alpha V_0 (\xi_w - \ln 2) + c \left(\frac{\alpha}{V_0} \right)^{1/3}, \quad c = 1.0 \pm 0.1 \quad (9.2)$$

(c absorbs the residual quantization offset; the empirical fit across 11 series gives c in 0.85–1.0). Both terms are *independent of ε_s* : the shear width does not select the wavelength — the coupling and the containment radius select it together. Verified to ± 0.6 in k_z on all 11 usable series, including the direct window test: moving the cut $11 \rightarrow 25$ at fixed physics ($\alpha = 2$, $V_0 = 0.05$) moved the measured peak $4.5 \rightarrow 5.5$ (formula: $4.45 \rightarrow 5.85$).

9.3 Beyond-peak roll-off

Past the peak, model A applies with the window-truncated well. Two limits of (8.11):

Step 9.3.1 (Loose-window limit $X \rightarrow 0$). $1 + X^2 \rightarrow 1$ gives the envelope of the truncated branch,

$$\gamma^2 = \frac{1}{2} \left[-V_0^2 k_z^2 + \sqrt{V_0^4 k_z^4 + 8\alpha V_0^3 k_z} \right]. \quad (9.3)$$

Step 9.3.2 (High- k_z asymptotics). For $8\alpha/(V_0 k_z^3) \ll 1$ expand the square root, $\sqrt{1 + \epsilon} \simeq 1 + \epsilon/2$ with $\epsilon = 8\alpha/(V_0 k_z^3)$:

$$\gamma^2 \simeq \frac{1}{2} V_0^2 k_z^2 \cdot \frac{4\alpha}{V_0 k_z^3} = \frac{2\alpha V_0}{k_z} \quad \Rightarrow \quad \gamma \simeq \sqrt{\frac{2\alpha V_0}{k_z}} \quad (9.4)$$

— the measured beyond-peak decline ($\gamma \propto k_z^{-1/2}$), steepened further by the $(1 + X^2)^{-1}$ quantization factor as the truncated well narrows.

9.4 Why the data said $k_z^{\text{peak}} \approx 2\alpha$; γ^{peak}

Over the surveyed grid ($\alpha = 1\text{--}5$, $V_0 = 0.03\text{--}0.1$, window radii shrinking as α grew, roughly $\alpha V_0 \xi_w \approx 1$), formula (9.2) numerically tracks $\approx 2\alpha$: the sub-linear resonance term $(\alpha/V_0)^{1/3}$ plus a window term whose radius the campaign design shrank as α grew mimics a linear trend across a factor-5 range of α . The “ $k_z^{\text{peak}} \approx 2\alpha$ law” is a serviceable survey-box mnemonic; (9.2) is the law. Similarly, the empirical $\gamma^{\text{peak}} \sim V_0^{0.85\text{--}0.92}$ at fixed α is the ceiling’s $V_0^{2/3}$ times the window/quantization correction $\hat{\gamma}(\text{budget}, \alpha V_0 \xi_w)$, which increases with V_0 across the surveyed range; $\gamma^{\text{peak}} = (0.95\text{--}0.99)\Gamma$ for $\xi_w \geq 8$, dropping to $\sim 0.95\Gamma$ at $\xi_w = 5$.

10 Shear-width scaling: the quantization budget

How does ε_s enter, given that neither the ceiling (7.4) nor the peak formula (9.2) contains it? Answer: only through the *action*, i.e. the quantization budget. We nondimensionalize the exact action integral step by step. (This underlies the EPS-scan v2 campaign design and its Phase-0 theory, `analysis/eps_collapse_theory.py`.)

Step 10.0.1 (Exact action in the w variable). In the saturated region, $u = \alpha V_0(|\xi| - \ln 2)$ and $x = \varepsilon_s \xi$ give $dx = \varepsilon_s d\xi = \frac{\varepsilon_s}{\alpha V_0} du$, and $w = k_z - u$ makes $|dw| = |du|$:

$$S = \int_{\text{well}} \sqrt{Q} dx = \frac{\varepsilon_s}{\alpha V_0} \int_{\text{well}} \sqrt{Q} dw. \quad (10.1)$$

Step 10.0.2 (Hatted integrand). With $w = K\hat{w}$ and (7.6),

$$\mathcal{D} = \gamma^2 + \Omega_A \Omega_F = \gamma^2 + w(2k_z - w) = K^2 \left[\hat{w}(2\hat{k} - \hat{w}) + V_0^2 \hat{\gamma}^2 \right] \equiv K^2 \hat{\mathcal{D}}, \quad (10.2)$$

$$Q = \frac{\mathcal{D}(G - \gamma^2)}{\gamma^2} = K^2 \frac{\hat{\mathcal{D}}(\hat{G} - \hat{\gamma}^2)}{\hat{\gamma}^2} \equiv K^2 \hat{Q}(\hat{w}; \hat{k}, \hat{\gamma}) \quad (10.3)$$

(the factors Γ^2 from $G - \gamma^2$ and γ^2 cancel; the explicit $V_0^2 \hat{\gamma}^2 \leq 0.04$ inside $\hat{\mathcal{D}}$ is negligible over the production range, so $\hat{Q}$ is universal in hatted variables).

Step 10.0.3 (Assembled scaling). Using $dw = K d\hat{w}$ together with

$$\frac{K^2}{\alpha V_0} = \left(\frac{\alpha}{V_0} \right)^{2/3} \frac{1}{\alpha V_0} = \alpha^{-1/3} V_0^{-5/3} = \frac{1}{V_0 (\alpha V_0^2)^{1/3}}, \quad (10.4)$$

we obtain

$$S = \frac{\varepsilon_s K^2}{\alpha V_0} \int \sqrt{\hat{Q}} d\hat{w} = \frac{\varepsilon_s}{V_0 (\alpha V_0^2)^{1/3}} \hat{I}(\hat{k}, \hat{\gamma}), \quad \hat{I} \equiv \int_{\hat{w}-\text{well}} \sqrt{\hat{Q}} d\hat{w} \quad (10.5)$$

(numerically verified: at $\alpha = 2$, $V_0 = 0.05$, $\varepsilon_s = 0.15$, $k_z = 4$, $\gamma = 0.16$, both sides give $S \approx 2.8$ before quantization pulls γ up). The shear width therefore enters the eigenvalue problem *only* as a multiplicative budget on the action; the well's shape in $\hat{w}$ is ε_s -free.

Remark 10.1 (Normalization of the program's "P" group). The project's shorthand (PHYSICS.md eq. (11.5)) writes $S = (\varepsilon_s/\Gamma) \int \sqrt{\hat{Q}'} d\hat{w}$ with the group $P = \Gamma/\varepsilon_s$; comparing with (10.5), that convention absorbs a factor V_0^{-2} into its $\hat{Q}'$ ($\hat{Q}' = \hat{Q}/V_0^2$). The clean statement is (10.5): the dimensionless budget prefactor is $\varepsilon_s^{-1} V_0 (\alpha V_0^2)^{1/3} = (\alpha V_0^5)^{1/3} / \varepsilon_s$, with the integral $\hat{I}$ universal. Either bookkeeping gives identical model-A/B numbers (the campaign tooling evaluates the models directly and never uses the shorthand); the distinction matters only if one collapses *data* against a single "budget group", where the residual V_0 -dependence is part of why a pure one-group collapse in $P = (\alpha V_0^2)^{1/3} / \varepsilon_s$ is imperfect (cf. the observed ε_s -drift of k_z^{peak} at fixed window).

Remark 10.2 (Qualitative consequences). Larger ε_s (at fixed window and physics) enlarges S at given γ , and since S decreases in γ , quantization then pushes γ *up* toward the ceiling — the sign of the measured 10–39% rise of γ from $\varepsilon_s = 0.10 \rightarrow 0.45$ at fixed $(k_z, \xi_{\text{sponge}})$. Because the budget also shifts which k_z first saturates its quantization, k_z^{peak} acquires a mild ε_s -dependence even though (9.2)'s leading terms have none.

11 Diagnosis of the WKB quartic's domain of validity

The quartic (2.4) models the drive as the constant $-\alpha^2 V_0 k_z / \omega^2$ term of (1.3) — one power of v (from the current), one resonance-free factor $\alpha^2 V_0 k_z$. The true drive (7.1) has three powers of v , is Doppler-resonant, and is ceiling-bounded by (7.3). Comparing the two fixes exactly where eq. (1.4)/(2.4) can and cannot work:

- **Low k_z (below the true peak):** the quartic's drive $\alpha^2 V_0 k_z / \gamma^2$ has no ceiling; as γ is iterated it can supply arbitrarily large growth, and the quartic overshoots the true branch by the measured 10–20 $\times$ in the $k_z \lesssim 1$ tail.
- **High k_z :** the quartic's balance $\omega^4 - k_z^2 \omega^2 \approx \alpha^2 V_0 k_z$ makes $\gamma^2 \approx \alpha^2 V_0 / k_z$, a *decline*, while the true unbounded branch saturates at the ceiling; in windowed data the truncated true branch declines as (9.4) instead. In the weak-coupling corner this produces the observed $\sim 25\%$ quartic underestimates.
- **$k_z = 0$:** the geometry changes (parabolic potential exact, different channel); the harmonic-core quantization is exact there and the 0.5% chromo-Weibel anchor (2.8) stands untouched by this critique.

The quartic is thus a valid *harmonic-core* theory (its own regime: $k_z = 0$ and the immediate neighborhood of a parabolic well) but not a theory of the finite- k_z shear branch; models A/B replace it there.

12 Cold two-stream rates in code units

The counter-streaming beams support, away from the shear layer ($v_z \rightarrow \pm V_0$), an electrostatic color-1 two-stream instability. Its exact cold rates in code units calibrate the bandpass-filter design (Appendix B) and were used to test (and reject) the outer two-stream as the late-catastrophe driver.

12.1 Dispersion relation

Step 12.1.1. Longitudinal channel: perturbations $\propto e^{i(kz - \omega t)}$ of n_s, v_{zs} , coupled by the longitudinal field E_z . Linearized continuity and momentum for beam s with background velocity u_s (here $u_{A,B} = \pm V$), per-beam plasma frequency (1.5):

$$\delta n_s = \frac{k n_0 \delta v_s}{\omega - k u_s}, \quad \delta v_s = \frac{i q_s E_z / m}{-(i)(\omega - k u_s)} \implies \delta n_s = -\frac{n_0 q_s E_z k / m}{(\omega - k u_s)^2} \cdot \frac{1}{i} \cdot i = -\frac{k q_s n_0 E_z / m}{(\omega - k u_s)^2}. \quad (12.1)$$

Step 12.1.2. Poisson/Gauss ($\varepsilon_0 = 1$): $ikE_z = \sum_s q_s \delta n_s$ gives

$$ikE_z = -\sum_s \frac{k q_s^2 n_0 E_z / m}{(\omega - k u_s)^2} \cdot \frac{1}{i^{-1}} \implies 1 = \sum_s \frac{\omega_p^2}{(\omega - k u_s)^2} = \frac{1}{(\omega - kV)^2} + \frac{1}{(\omega + kV)^2}. \quad (12.2)$$

(the standard two-beam form; each beam contributes $\omega_p^2 = 1$).

12.2 Closed-form growth rate

Step 12.2.1 (Reduce to a quadratic). Put the right side of (12.2) over a common denominator, using $(\omega \mp kV)^2$ sum = $2(\omega^2 + k^2 V^2)$:

$$(\omega^2 - k^2 V^2)^2 = 2(\omega^2 + k^2 V^2). \quad (12.3)$$

Let $y = \omega^2$ and $\beta = k^2 V^2$:

$$(y - \beta)^2 = 2(y + \beta) \iff y^2 - 2(\beta + 1)y + \beta^2 - 2\beta = 0. \quad (12.4)$$

Step 12.2.2 (Solve).

$$y_{\pm} = (\beta + 1) \pm \sqrt{(\beta + 1)^2 - \beta^2 + 2\beta} = (\beta + 1) \pm \sqrt{4\beta + 1}. \quad (12.5)$$

Step 12.2.3 (Instability condition). $y_- < 0$ (i.e. $\omega^2 < 0$, purely growing) iff $(\beta + 1)^2 < 4\beta + 1$ iff $\beta^2 < 2\beta$ iff $\beta < 2$:

$$\boxed{kV < \sqrt{2}} \quad \text{unstable band.} \quad (12.6)$$

At $V_0 = 0.1$ this is $k < 14.1$ — exactly the documented $k_{ts} \approx 14$ two-stream band that fixed the production bandpass edge.

Step 12.2.4 (Growth rate, maximum, small- k limit).

$$\boxed{\gamma^2(kV) = -y_- = \sqrt{4(kV)^2 + 1} - (kV)^2 - 1} \quad (12.7)$$

Maximize over β : $d\gamma^2/d\beta = 2/\sqrt{4\beta + 1} - 1 = 0$ gives $\beta = 3/4$, i.e.

$$kV = \frac{\sqrt{3}}{2}, \quad \gamma_{\max}^2 = \sqrt{4} - \frac{3}{4} - 1 = \frac{1}{4}, \quad \gamma_{\max} = \frac{1}{2}. \quad (12.8)$$

Small kV : $\sqrt{4\beta + 1} \simeq 1 + 2\beta - 2\beta^2$ gives $\gamma^2 \simeq \beta$, i.e. $\gamma \simeq kV$. (Measured full-domain rates 0.7–0.9 exceed $\frac{1}{2}$ because they include the electromagnetic/Weibel corrections outside this electrostatic model.)

13 Nonlinear theory: cavitation threshold and the back-reaction law

13.1 Density cavitation of the finite-amplitude KH mode

The linear eigensolver state vector contains no fluid density/momentum blocks; the late-time “hard wall” at $V_0 \gtrsim 0.08$ is a nonlinear fluid effect. Its threshold is derived as follows.

Step 13.1.1 (First-order compressive response). Linearize (3.1) about $n = 1$ for a mode $\propto e^{\gamma t}$:

$$\gamma \delta n = -(\mathrm{i}k_z \delta p_z + \partial_x \delta p_x) \quad \implies \quad |\delta n| \sim \frac{k_z |p_{z1}| + k_x^{\mathrm{eff}} |p_{x1}|}{\gamma}, \quad (13.1)$$

where k_x^{eff} is the mode’s inverse x -scale.

Step 13.1.2 (Second-order (ponderomotive) part). Quadratic beats of the mode grow at 2γ and accumulate a z -mean density modification; the observed density-perturbation growth at the retained k_z at twice the saturating field rate is this quadratic part taking over as the fields saturate.

Step 13.1.3 (Cavitation threshold). Cavitation ($\delta n \rightarrow n_0$, i.e. n hits the code floor 0.05) occurs when the momentum amplitude reaches

$$\boxed{p_{\mathrm{cav}} \sim \frac{\gamma}{\max(k_z, k_x^{\mathrm{eff}})}} \quad (13.2)$$

(order of magnitude). Beyond it, the floored divisions $v = p/\max(n, 0.05)$ decouple momentum from density and the model no longer represents the physical system; a long-lived floored-pocket state persists for tens of TU before terminal blow-up. Because (13.2) is an *amplitude* threshold, whether a run cavitates within its length is controlled by drive strength (V_0 , through γ and the saturation level) and effective seed — hence the observed gradual $V_0 = 0.05 \rightarrow 0.10$ transition, its insensitivity to window mechanism and radius, and the practical rule of capping `target_tu` just past the plateau at high V_0 . Forensics on a failing run localized every growing channel at the shear layer (99.9% of density-perturbation power at $|\xi| < 5$) with the outer region quiet, ruling out the outer branches as the driver.

13.2 Structure of the back-reaction on A_{z1} : the quadratic law

Unfreezing A_z^1 lets the color-2/3 wave imprint on the background. The measured law (two independent configurations, four decades of $|a|^2$, endpoint exact) is

$$\Delta A_{z1} \simeq -0.04 |a|^2, \quad (13.3)$$

localized at the mode peak, i.e. the wave *deepens* $|A_{z1}|$ — opposite in sign to depletion. The structure (quadratic, $k_z = 0$, localized, growing at 2γ) is derived as follows; the coefficient and sign are taken from measurement.

Step 13.2.1 (All color-1 second-order sources are helicity beats). For real fields with circular structure $X^2 = \text{Re}(\hat{X}e^{ik_z z})$, $X^3 = \text{Im}(\hat{X}e^{ik_z z})$, any color-1 cross-product source is a beat of two color-2/3 waves, e.g. for the precession of Q^1 :

$$(\vec{Q} \times \vec{A}_z)^1 = Q^2 A_z^3 - Q^3 A_z^2 = \text{Re}(\hat{q}e^{i\theta}) \text{Im}(\hat{a}e^{i\theta}) - \text{Im}(\hat{q}e^{i\theta}) \text{Re}(\hat{a}e^{i\theta}) = \text{Im}(\hat{a} \hat{q}^*), \quad \theta = k_z z, \quad (13.4)$$

which is *independent of z* : the color-1 imprint of a circular mode is purely $k_z = 0$, and $\propto |\text{amplitudes}|^2 e^{2\gamma t}$. The same computation applies to the current beats ($\delta n \delta v_z$, $\delta Q^{2,3} \delta v_z$ contributions to J_z^1) and to the field commutator beats in (3.5)/(3.7).

Step 13.2.2 (Chain to ΔA_{z1}). The $k_z = 0$ color-1 source drives $J_z^{1(2)} \rightarrow E_z^1$ (Ampère- z), and $\partial_t A_{z1} = -E_z^1$ integrates it. With every second-order source $\propto e^{2\gamma t}$, each time integration divides by 2γ , so

$$\Delta A_{z1}(t) = -\frac{(\text{source strength})}{(2\gamma)^2} e^{2\gamma t} \propto -|a(t)|^2, \quad (13.5)$$

a pure quadratic law in the instantaneous mode amplitude, pinned at the mode peak — exactly the measured structure (13.3).

Step 13.2.3 (Consequences). Since $\Delta A_{z1} < 0$ where $A_{z1} < 0$, $|\alpha A_{z1}|$ grows: no depletion of the battery. The clean frozen/unfrozen pair shows the global effect is weakly stabilizing (mode amplitude -7% by $|a| \approx 2.5$; annulus compression against the fixed damping boundary), two orders of magnitude too weak to saturate the branch: saturation is by fluid nonlinearity (§13.1), not background drain, and the frozen- A_{z1} approximation is accurate to a few percent everywhere it matters (validated: unfrozen linear rate unchanged, $\gamma = 1.45$ vs eigensolver 1.43–1.46 at the reference blow-up).

14 Warm isothermal fluid closure

14.1 Derivation of the pressure source

Take the first velocity moment of the kinetic equation for beam s with an isotropic thermal spread T (code units, $m = 1$):

$$\partial_t p_i + \partial_j (v_j p_i) = -\partial_i P + F_i, \quad P = nT \quad (\text{isothermal closure}), \quad (14.1)$$

so the only new term relative to the cold model (3.2)–(3.3) is the momentum-density source

$$\boxed{\delta F_i = -T \partial_i n} \quad (14.2)$$

(no extra $1/n$: the sources are momentum-density rates). $T = 0$ recovers the cold system identically. Linearly, the closure adds the sound term $c_s^2 = T$ to the fluid response: the two-stream channel of §12 acquires the standard thermal stabilization when $v_{\text{th}} = \sqrt{T} \gtrsim V_0$ (Landau-fluid analogue of $kV < \sqrt{2}\omega_p$ shrinking), which was the intended validation target.

14.2 Why it cannot stabilize the $k_z = 0$ chromo-Weibel

The $k_z = 0$ mode of §2.2 is a filamentation (Weibel-class) instability driven by the momentum-space *anisotropy* of the counter-streaming beams. An isotropic scalar pressure $P = nT$ adds a restoring force only along density gradients of the compressive response; it does not relax the anisotropy that powers filament growth. The scan confirmed this: across $v_{\text{th}}/V_0 = 0\text{--}3$ the final $k_z = 0$ amplitude changed by only $\sim 11\%$. (The clean warm-vs-cold validation was subsequently obtained on the KH target point itself: with the corrected control, warm-closure rates match the cold theory at the 1% level, closing T1.4.)

15 Eigenmode seeding theory

15.1 Hermite ground state of the harmonic core

The harmonic core of the given theory, $\partial_x^2 a + (Q_0 - |C_1|x^2)a = 0$ (from (1.2)), is exactly the quantum harmonic oscillator: with $\sigma \equiv |C_1|^{-1/4}$, eigenfunctions and eigenvalues are

$$a_n(x) = H_n\left(\frac{x}{\sigma}\right) e^{-x^2/2\sigma^2}, \quad Q_0 = (2n+1)\sqrt{|C_1|}, \quad (15.1)$$

i.e. eq. (1.4) is the oscillator spectrum, and the $n = 0$ eigenfunction is the Gaussian

$$a_0(x) = \exp\left(-\frac{x^2}{2\xi_{\text{char}}^2}\right), \quad \xi_{\text{char}} = |C_1|^{-1/4} = \left(\frac{2\gamma^2}{\alpha^3 V_0^2}\right)^{1/4} \quad (15.2)$$

(using $|C_1| = \alpha^3 V_0^2/(2\omega^2)$, $\omega^2 \rightarrow -\gamma^2$ with the magnitude convention of §2.1; x here is the ξ -coordinate of the step-potential form). With the high- k_z quartic balance $\gamma^2 \approx \alpha^2 V_0/k_z$ (§11),

$$|C_1| \approx \frac{\alpha V_0 k_z}{2} \implies \xi_{\text{char}} \approx \left(\frac{2}{\alpha V_0 k_z}\right)^{1/4}. \quad (15.3)$$

Remark 15.1 (The implemented width). Campaign 18's init kernel uses the simpler $\xi_{\text{char}}^{\text{code}} = (\alpha k_z V_0)^{-1/2}$, which coincides with (15.3) at $\alpha V_0 k_z \approx 1.4$ and tracks it within a few tens of percent over the campaign grid (e.g. $\alpha = 2$, $V_0 = 0.1$, $k_z = 2$: code 1.58 vs Hermite ≈ 1.5). A width mismatch only launches a transient that decays within a few growth times, so the approximation is operationally adequate; the exact profile is superseded anyway by full six-field seeding (below).

15.2 Why single-field seeding selects wrong modes

An A_z -only (or B_y -only) seed fixes one field's profile and lets the activation chain build the others; the projection of that initial condition onto the eigenmode basis decides what grows first. Concretely (C23): for the target mode at $(\alpha, V_0, k_z) = (1, 0.05, 2)$ the B_y and A_z profiles peak at different ξ (1.64 vs 7.85); an A_z -only Gaussian placed correct-for- A_z deposits the charge response at the wrong location, and a different (co-located, slower) eigenmode wins the race: measured $\gamma = 0.060$ vs target 0.122. The remedy is seeding all six fields with the solver eigenvector (exact amplitude ratios and phases), which removes the transient entirely; conversely §8.5 shows the seed must come from the *dominant* (σ -chased) eigenvector, or the run will faithfully measure an overtone.

15.3 Cascade race condition

Single-field B_y seeding at $k_z \geq 2$ loses a second race: the layer-scale color mixing of §5.2 builds $A_z^{2,3}$ at $\gamma_{\text{casc}} \approx \alpha V_0$ from any residual field, so if $\gamma_{\text{KH}} < \gamma_{\text{casc}}$ the seed decays into the cascade background before the KH chain closes. Gaussian A_z seeding (mode 6) pre-loads the slowest link of the chain $a \rightarrow q_A \rightarrow \text{Lorentz} \rightarrow b$ at the correct width (15.2), which is why it (and its six-field successor) restored clean linear windows at $k_z \geq 2$.

A Helicity projections and the conjugate-extraction identity

For a circular mode the two real fields are

$$A_z^2(x, z) = \text{Re}[\hat{a}(x)e^{ik_z z}], \quad A_z^3(x, z) = \text{Im}[\hat{a}(x)e^{ik_z z}], \quad (\text{A.1})$$

so the complex pair is single-signed in k_z : $a = A_z^2 + iA_z^3 = \hat{a}(x)e^{ik_z z}$ exactly. Its z -DFT therefore has *all* its weight at $+k_z$ and none at $-k_z$:

$$\tilde{a}(+k_z) = \hat{a}, \quad \tilde{a}(-k_z) = 0. \quad (\text{A.2})$$

Writing the projections in real arithmetic with $c = \cos k_z z$, $s = \sin k_z z$:

$$\tilde{a}(\pm k_z) \propto \sum_z [(A_z^2 c \pm A_z^3 s) + i(\mp A_z^2 s + A_z^3 c)]. \quad (\text{A.3})$$

The extraction bug of 2026-07-04→07-17 computed the lower sign (the $-k_z$, conjugate-helicity projection): by (A.2) this is *identically zero* for a pure circular mode, and what it actually returned on real data was the small opposite-helicity contamination — hence amplitudes at the e^{-25} floor and the triage rule “first-amplitude $< e^{-25} \Rightarrow$ invalid extraction”. The mirror symmetry of §4.3 guarantees the physical information lives in one helicity per side of the layer, so the corrected single-sign projection loses nothing.

B Filters, sponges, and hyperdiffusion as linear operators

DFT band filter. `kernel_ym_subtract_kz_range` applies, each step, the projection $f \mapsto f - \sum_{k \in \mathcal{B}} \Pi_k f$, where Π_k is the exact DFT projector onto z -mode k and $\mathcal{B} = \{1, \dots, k_{\text{lo}}\} \cup \{k_{\text{mode}} + 1, \dots, k_{\text{hi}}\}$: an idempotent linear operator that removes the two-stream band (12.6) while leaving the measured mode untouched. The same projector acts on the fluid p_z ’s. (Empirical constraint, recorded not derived: keep $k_{\text{hi}} \leq N_Z/4.5$ so the projector’s stencil does not interact with the FCT limiter’s near-Nyquist deposits.)

Sponge. `kernel_ym_sponge` adds $\partial_t f = -\sigma_s(\xi)f$ with $\sigma_s = \sigma(|\xi| - \xi_{\text{sponge}})_+$: a non-Hermitian absorbing potential. In the eigenproblem it appears as $-\sigma_s(\xi)$ on the diagonal of all blocks; it removes the outer branch of §5.2 when $\xi_{\text{sponge}} < \xi_{\text{crit}}$ (legitimate, by the battery argument) at the cost of the few-percent well compression measured in the sponge-vs-cut comparisons.

Hard cut. `xi_cut` zeroes all wave fields outside ξ_w : Dirichlet/Neumann wall pair of §6.3.

Hyperdiffusion. The z -hyperdiffusion adds $\partial_t f = -\mu \partial_z^4 f$, i.e. damping μk^4 per mode: with $\mu = 5 \times 10^{-5}$ it removes grid-scale noise while its drag on a retained mode k is $\mu k^4 \ll \gamma$ for all production k (e.g. $k = 6$: $\mu k^4 = 0.065\gamma$ at $\gamma = 0.1$; the design condition is $1/(\mu k^4) \gg 1/\gamma$).

C Validation ledger

Derived result	Numerical anchor
Quartic $k_z=0$ chromo-Weibel rate (2.8)	sim vs formula: 0.5%; $\gamma = 0.284$ reproduced at the $\alpha = 1$, $V_0 = 0.05$ reference
Six-field system (4.17)	eigensolver vs 2D CUDA growth rates: 1–4% (plateau-confirmed clean cores: median 6–10% archive-wide)
Exact reduction (6.5)/(6.8)	reduced operator vs 6-field ARPACK: rel. diff. 0.0 ($< 10^{-10}$) at 7 points, $\alpha = 1-3$, $V_0 = 0.03-0.1$, $k_z = 1-7$
Tachyonic law (5.4)–(5.5)	analytic envelope = float64 eigensolver = float32 CUDA rates at all outer eigenmode peaks; blow-up $\gamma = 1.45$ vs 1.43–1.46
Ceiling (7.4)	dominant-branch peaks at 95–99% of $(\alpha V_0^2)^{1/3}$ in every series; T2.5 collapse $\gamma_{\text{peak}}/\Gamma = 0.977 \pm 0.011$
Model B quantization (§8.4)	$\leq 2\%$ (median $\leq 0.2\%$) vs σ -chased $n=0$ eigenvalues, 12 series, $k_z \geq 1.5$
Model A closed form (8.11)	$\sim 3\%$ near/above peak; peak located to ± 0.5 in k_z
Peak formula (9.2)	± 0.6 in k_z across 11 series; window test: cut 11 $\rightarrow$ 25 moved peak 4.5 $\rightarrow$ 5.5 (formula 4.45 $\rightarrow$ 5.85)
Two-stream band (12.6)	$k_{\text{ts}} \approx 14$ at $V_0 = 0.1$ (documented band edge)
Back-reaction law (13.3)	$\Delta A_{z1} = -0.04 a ^2$ over four decades; endpoint -0.242 vs -0.242
Warm closure (§14)	warm vs cold theory at the KH target: 1% (corrected control)